

Package ‘clis’

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Type Package

Title Conformal Local Influence Screening for Bounded-Response Regression

Version 0.3.6

Description Provides fast, statistically calibrated influence diagnostics for zero-or-one inflated beta (BIC) regression models with variable dispersion. The core idea is to use the conformal normal curvature of Poon and Poon (1999) <doi:10.1111/1467-9868.00162> as a non-conformity score within a split-conformal testing procedure, yielding per-observation conformal p-values whose Benjamini-Hochberg adjustment controls the false discovery rate at a user-specified level (Bates and others, 2023) <doi:10.1214/22-AOS2244>. Unlike classical local influence diagnostics, which rely on visual inspection of index plots and do not scale beyond a few hundred observations, 'clis' provides a finite-sample error guarantee and runs in linear time per observation after a single model fit. Methods for four perturbation schemes, block decomposition of influence into the inflation-probability and conditional-mean/precision components, penalised additive (semiparametric) submodels, and a full suite of diagnostic plots are included.

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URL <https://github.com/Raydonal/clis>, <https://raydonal.github.io/clis/>

BugReports <https://github.com/Raydonal/clis/issues>

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bic_info	<i>Observed or Fisher information matrix for a BIC model</i>
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Description

Computes the block-structured information matrix of a fitted zero-or-one inflated beta regression with variable dispersion. By construction the matrix is block diagonal between the inflation parameters and the mean/precision parameters (information orthogonality).

Usage

```
bic_info(object, use_fisher = TRUE, penalty = NULL)
```

Arguments

object	A fitted <code>gamlss</code> model of family BEZI or BEOI.
use_fisher	Logical; if TRUE (default) returns the expected (Fisher) information, which is guaranteed positive definite; if FALSE returns the observed information.
penalty	Optional penalty matrix S of dimension $(M+m+p) \times (M+m+p)$ for penalised additive submodels. When supplied, the returned information is the penalised information $J + S$ (or $I + S$), following the semiparametric extension. The penalty must be block diagonal across the gamma, beta, delta groups so that separability is preserved; see <code>bic_penalty()</code> .

Value

A list with the information matrix (`info`), its inverse (`info_inv`), the parameter-block indices (`idx`), the weight sequences (`weights`), and (if a penalty was supplied) the effective degrees of freedom (`edf`) and their block split (`edf_blocks`).

References

Ospina, R. and Ferrari, S. L. P. (2012). A general class of zero-or-one inflated beta regression models. *Computational Statistics & Data Analysis*, 56(6), 1609-1623.

bic_link

*Link function and its derivatives for Blc submodels***Description**

Constructs a list with the link function, its inverse, and the first and second derivatives of the inverse link, used throughout the influence computations.

Usage

```
bic_link(link)
```

Arguments

link	Character string naming the link. One of "logit", "probit", "cloglog", or "log".
------	--

Value

A list with components `name`, `linkfun`, `linkinv`, `mu.eta` (first derivative of the inverse link), and `mu.eta2` (second derivative).

Examples

```
lk <- bic_link("logit")
lk$mu.eta(0.3) # d mu / d eta at mu = 0.3
```

bic_penalty	<i>Extract the penalty matrix from a fitted additive Blc model</i>
-------------	--

Description

Assembles the block-diagonal penalty matrix $S(\lambda)$ from the smooth terms of a fitted `gamlss` model whose submodels use penalised additive terms (for example `pb()` P-spline terms). The result is suitable as the `penalty` argument of `bic_info()`.

Usage

```
bic_penalty(object)
```

Arguments

<code>object</code>	A fitted <code>gamlss</code> model of family <code>BEZI</code> or <code>BEOI</code> with penalised additive terms in one or more submodels.
---------------------	---

Details

The function reads the smoothing structure stored by `gamlss` for each parameter (`mu`, `sigma`, `nu`) and places the corresponding penalty contributions on the diagonal blocks. Terms without a penalty contribute a zero block, so a model with a mix of linear and smooth terms is handled transparently. By construction the returned matrix is block diagonal across the three coefficient groups, so it preserves the separability required by the semiparametric theory.

Value

A block-diagonal penalty matrix of dimension $(M+m+p) \times (M+m+p)$, block diagonal across the inflation, mean, and precision coefficient groups.

See Also

[bic_info\(\)](#)

bic_quantile_residuals	<i>Randomized quantile residuals for BEZI/BEOI models</i>
------------------------	---

Description

Computes randomized quantile residuals for a fitted zero- or one-inflated beta model. For the continuous observations the beta CDF is scaled by the non-inflation probability; for the boundary observations the residual is randomized uniformly over the probability atom, which is the standard construction for a mixed discrete-continuous distribution. This is provided because the generic `residuals()` method does not always handle the inflation atom correctly, which can make every residual share the same sign.

Usage

```
bic_quantile_residuals(object, seed = NULL)
```

Arguments

object A fitted `gamlss` model of family BEZI or BEOI.
seed Optional integer seed for the randomization at the boundary.

Value

A numeric vector of randomized quantile residuals.

clis_screen	<i>Conformal Local Influence Screening</i>
-------------	--

Description

Performs scalable, FDR-controlled influence screening for a fitted zero-or-one inflated beta regression model. The conformal normal curvature score of each observation is used as a non-conformity score within a split-conformal procedure: the data are partitioned into a calibration set (presumed clean) and a screening set; conformal p-values are computed for the screening set; and Benjamini-Hochberg adjustment declares an influential subset with false discovery rate controlled at level α .

Usage

```
clis_screen(
  object,
  scheme = "caseweights",
  p = 1L,
  alpha = 0.1,
  calib_frac = 0.5,
  calib_idx = NULL,
  use_fisher = TRUE,
  r_max = 4L,
  score = c("B_Et", "m_r"),
  penalised = FALSE,
  penalty = NULL,
  seed = NULL
)
```

Arguments

object A fitted `gamlss` model of family BEZI or BEOI.
scheme Character; the perturbation scheme. One of "caseweights" (default), "disccovar", "meancovar", or "preccovar".

<code>p</code>	Integer; covariate index for the covariate-perturbation schemes.
<code>alpha</code>	Numeric in (0, 1); target false discovery rate. Default 0.1.
<code>calib_frac</code>	Numeric in (0, 1); fraction of observations used for conformal calibration when <code>calib_idx</code> is not supplied. Default 0.5.
<code>calib_idx</code>	Optional integer vector of observation indices to use as the calibration set. The conformal false discovery rate guarantee requires the calibration set to be (nearly) free of influential points; when a trusted clean subset is known, supply it here. If NULL (default) a random subset of size $\text{floor}(\text{calib_frac} * n)$ is drawn, which is appropriate only when influential points are rare.
<code>use_fisher</code>	Logical; use the Fisher information (default TRUE).
<code>r_max</code>	Integer; order for aggregate contributions used as the score.
<code>score</code>	Character; which CNC quantity to use as the non-conformity score: "B_Et" (basic perturbation, default) or "m_r" (aggregate contribution at order <code>r_max</code>).
<code>penalised</code>	Logical; if TRUE, use the penalised information $J + S$ obtained from the smooth terms of an additive <code>gamLSS</code> fit, implementing the semiparametric extension. The penalty is extracted with <code>bic_penalty()</code> . Default FALSE.
<code>penalty</code>	Optional penalty matrix, of the dimension of the full coefficient vector, added to the information before inversion. Use this when the smooth basis is supplied explicitly as design columns.
<code>seed</code>	Optional integer for reproducible calibration splitting.

Details

Classical local influence diagnostics rank observations by a curvature measure and rely on visual inspection of an index plot against an ad-hoc threshold. This neither scales to large n nor provides any control of the type-I error rate. CLIS addresses both limitations: the conformal wrapper supplies a finite-sample FDR guarantee, and the per-observation score is computed in linear time after a single model fit.

The calibration set is assumed to be predominantly free of influential observations. Because influential points are typically rare, a random split satisfies this approximately; for adversarial settings, a robust pre-filter can be applied before calibration.

Value

An object of class `clis` with components:

influential Integer indices (into the screening set) declared influential at FDR level `alpha`.

influential_global Integer indices into the original data.

pvalues Conformal p-values for the screening set.

padj Benjamini-Hochberg adjusted p-values.

scores Non-conformity scores for all observations.

calib_idx, screen_idx Calibration and screening indices.

alpha, scheme Inputs echoed back.

decomp Block decomposition (case-weights scheme only).

Examples

```

if (requireNamespace("gamlss", quietly = TRUE) &&
    requireNamespace("betareg", quietly = TRUE)) {
  data("ReadingSkills", package = "betareg")
  ReadingSkills$dys <- as.numeric(ReadingSkills$dyslexia == "yes")
  fit <- gamlss::gamlss(accuracy1 ~ dys * iq,
                        sigma.formula = ~ dys, nu.formula = ~ iq,
                        family = gamlss.dist::BEOI, data = ReadingSkills,
                        control = gamlss::gamlss.control(trace = FALSE))
  res <- clis_screen(fit, alpha = 0.1, seed = 1)
  print(res)
}

```

cnc_block_decomp

*Block decomposition of conformal normal curvature***Description**

Decomposes the per-observation CNC scores into a contribution from the inflation-probability sub-model and a contribution from the conditional-mean/precision submodel, exploiting the information orthogonality of the BIC model. The two contributions sum exactly to the total, with no cross terms.

Usage

```
cnc_block_decomp(delta_out, info_out)
```

Arguments

delta_out	A perturbation-matrix list (e.g. from delta_caseweights()) with a full Delta component.
info_out	An information list from bic_info() with info_inv and idx.

Value

A list with the discrete contribution B_gamma, the continuous contribution B_betadelta, their sum B_total, and the discrete fraction ratio_gamma.

cnc_matrix	<i>Conformal normal curvature matrix</i>
------------	--

Description

Computes the Frobenius-normalised conformal normal curvature (CNC) matrix of Poon and Poon (1999) for a given perturbation matrix and information matrix inverse.

Usage

```
cnc_matrix(Delta, info_inv)
```

Arguments

Delta	A perturbation matrix of dimension $(M+m+p) \times n$, e.g. from <code>delta_caseweights()</code> .
info_inv	The inverse information matrix from <code>bic_info()</code> .

Value

A list with the $n \times n$ CNC matrix B, its eigenvalues lambda (normalised so that the sum of squares is one), eigenvectors vectors, and the Frobenius norm normF of the unnormalised curvature.

References

Poon, W.-Y. and Poon, Y. S. (1999). Conformal normal curvature and assessment of local influence. *Journal of the Royal Statistical Society: Series B*, 61(1), 51-61.

cnc_scores	<i>Per-observation conformal normal curvature scores</i>
------------	--

Description

Computes the basic-perturbation CNC scores B_{E_t} and the aggregate contribution measures $m[r]_t$ from a fitted CNC matrix.

Usage

```
cnc_scores(cnc, r_max = 4L)
```

Arguments

cnc	A list returned by <code>cnc_matrix()</code> .
r_max	Integer; the maximum order of r-influential eigenvectors to consider for the aggregate contributions.

Value

A list with per-observation scores B_{Et} , the matrix of aggregate contributions m_r (one column per r), the eigenvalue thresholds $threshold_r$, the aggregate-contribution thresholds $threshold_{mt}$, the number of r -influential eigenvectors k_r , and the cutoff $b2$ for B_{Et} .

cnc_scores_linear	<i>Linear-time per-observation CNC scores</i>
-------------------	---

Description

Computes the basic-perturbation CNC scores B_{Et} without ever forming the $n \times n$ curvature matrix. The score is the scaled diagonal of $F_0 = \Delta^\top T^{-1} \Delta$, and both the diagonal and the Frobenius norm $\|F_0\|_F$ are obtained from quantities of dimension $(M + m + p)$, so the cost is linear in n and the memory footprint is negligible. This is the scalable path used for large samples, where the dense $n \times n$ eigenproblem of `cnc_matrix()` is infeasible.

Usage

```
cnc_scores_linear(Delta, info_inv)
```

Arguments

Delta	A perturbation matrix of dimension $(M+m+p) \times n$.
info_inv	The inverse information matrix from <code>bic_info()</code> .

Value

A list with per-observation scores B_{Et} , the cutoff $b2$, the Frobenius norm $normF$, and n .

conformal_pvalues	<i>Conformal p-values from non-conformity scores</i>
-------------------	--

Description

Given calibration scores (assumed to come from non-influential, "clean" observations) and test scores, computes marginal conformal p-values in the sense of Bates and others (2023). Larger scores indicate stronger evidence of influence, so the p-value for a test point is the calibrated rank of its score among the calibration scores.

Usage

```
conformal_pvalues(cal_scores, test_scores)
```

Arguments

cal_scores Numeric vector of calibration non-conformity scores.
 test_scores Numeric vector of test non-conformity scores.

Details

For a test score s and calibration scores $c_1, \dots, c_n$, the conformal p-value is

$$p = \frac{1 + \#\{i : c_i \geq s\}}{n + 1}.$$

These p-values are marginally valid (super-uniform under the null that the test point is exchangeable with the calibration set) and, by the positive-dependence result of Bates and others (2023), permit Benjamini-Hochberg FDR control.

Value

A numeric vector of conformal p-values, one per test score.

References

Bates, S., Candes, E., Lei, L., Romano, Y. and Sesia, M. (2023). Testing for outliers with conformal p-values. *The Annals of Statistics*, 51(1), 149-178.

Examples

```
set.seed(1)
cal <- rnorm(100)
test <- c(rnorm(8), 4, 5) # last two are outliers
conformal_pvalues(cal, test)
```

delta_caseweights	<i>Perturbation matrix for the case-weights scheme</i>
-------------------	--

Description

Perturbation matrix for the case-weights scheme

Usage

```
delta_caseweights(object)
```

Arguments

object A fitted BEZI/BEOI gamlss model.

Value

A list with the full Delta matrix and its gamma, beta, delta blocks.

delta_disccovar	<i>Perturbation matrix for the discrete-covariate scheme</i>
-----------------	--

Description

Perturbation matrix for the discrete-covariate scheme

Usage

```
delta_disccovar(object, p = 1L)
```

Arguments

object	A fitted BEZI/BEOI gamlss model.
p	Index of the perturbed covariate in the inflation design matrix.

Value

A list with the full Delta and its blocks.

delta_meancovar	<i>Perturbation matrix for the mean-covariate scheme</i>
-----------------	--

Description

Perturbation matrix for the mean-covariate scheme

Usage

```
delta_meancovar(object, p = 1L)
```

Arguments

object	A fitted BEZI/BEOI gamlss model.
p	Index of the perturbed covariate in the mean design matrix.

Value

A list with the full Delta and its blocks.

delta_preccovar	<i>Perturbation matrix for the precision-covariate scheme</i>
-----------------	---

Description

Perturbation matrix for the precision-covariate scheme

Usage

```
delta_preccovar(object, p = 1L)
```

Arguments

object	A fitted BEZI/BEOI gamlss model.
p	Index of the perturbed covariate in the precision design matrix.

Value

A list with the full Delta and its blocks.

envelope_bic	<i>Simulated envelope for quantile residuals</i>
--------------	--

Description

Simulated envelope for quantile residuals

Usage

```
envelope_bic(object, B = 200L, level = 0.95, seed = NULL)
```

Arguments

object	A fitted BEZI/BEOI gamlss model.
B	Integer; number of simulated samples.
level	Numeric; envelope coverage (default 0.95).
seed	Optional integer seed.

Value

Invisibly returns the observed residuals and envelope bands.

load_vaccination	<i>National DTP3 vaccination coverage (2022)</i>
------------------	--

Description

Reads country-level proportions of diphtheria-tetanus-pertussis (DTP3) vaccine coverage for 2022, with development covariates, from the CSV shipped in the package's extdata directory. The response places a point mass at one (countries achieving complete coverage), making it a natural application of one-inflated beta (BEOI) regression.

Usage

```
load_vaccination()
```

Value

A data frame with one row per country and the columns

iso3c ISO 3166-1 alpha-3 country code.

dtp3 DTP3 coverage proportion in $[0, 1]$ (response).

ln_gdp Natural log of GDP per capita (PPP, constant USD).

urb Urbanisation rate (proportion of urban population).

ln_pop Natural log of total population.

hdi Human Development Index in $[0, 1]$.

Source

WHO/UNICEF Joint Monitoring Programme (coverage); World Bank (GDP, urbanisation, population); UNDP Human Development Report (HDI).

Examples

```
vaccination <- load_vaccination()
summary(vaccination$dtp3)
mean(vaccination$dtp3 == 1) # fraction at the boundary
```

plot_clis	<i>Plot a CLIS screening result</i>
-----------	-------------------------------------

Description

Produces a two-panel diagnostic plot for a `clis_screen()` result: a plot of conformal p-values (with the Benjamini-Hochberg rejection boundary) and a plot of the non-conformity scores with the declared influential observations highlighted.

Usage

```
plot_clis(x, label_top = 8L, ...)
```

Arguments

<code>x</code>	A <code>clis</code> object from <code>clis_screen()</code> .
<code>label_top</code>	Integer; how many top influential points to label.
<code>...</code>	Currently ignored.

Value

Invisibly returns `x`.

plot_cnc_panels	<i>Plot the classical CNC diagnostic panels</i>
-----------------	---

Description

Plot the classical CNC diagnostic panels

Usage

```
plot_cnc_panels(
  cnc,
  scores,
  decomp = NULL,
  r_plot = 3L,
  label_top = 5L,
  suffix = ""
)
```

Arguments

cnc	A list from <code>cnc_matrix()</code> .
scores	A list from <code>cnc_scores()</code> .
decomp	Optional list from <code>cnc_block_decomp()</code> for a third panel.
r_plot	Integer; which aggregate-contribution order to plot.
label_top	Integer; number of points to label.
suffix	Character; appended to panel titles.

Value

Invisibly returns scores.

plot_influence	<i>Classical local-influence index plot</i>
----------------	---

Description

Reproduces the standard influence display of the beta-regression diagnostics literature: the per-observation conformal normal curvature B_{E_t} (or the aggregate contribution $m[r]_t$) plotted against the observation index, with the reference cutoff drawn and the most influential observations labelled. This is the plot practitioners of local influence expect to see; the conformal screening in `clis_screen()` and `plot_clis()` complements it with an error-controlled declaration.

Usage

```
plot_influence(
  object,
  scheme = "caseweights",
  p = 1L,
  measure = c("B_Et", "m_r"),
  r_plot = 3L,
  use_fisher = TRUE,
  penalised = FALSE,
  label_top = 5L,
  labels = NULL
)
```

Arguments

object	A fitted BEZI/BEOI <code>gamlss</code> model.
scheme	Perturbation scheme: "caseweights" (default), "disccovar", "meancovar", or "precovar".
p	Covariate index for the covariate-perturbation schemes.
measure	Which quantity to plot: "B_Et" (default) or "m_r".

r_plot	Aggregate-contribution order when measure = "m_r".
use_fisher	Logical; use the Fisher information (default TRUE).
penalised	Logical; use the penalised information (default FALSE).
label_top	Integer; number of most-influential points to label.
labels	Optional character vector of observation labels (for example country codes); defaults to the observation index.

Value

Invisibly, a list with the plotted scores and the cutoff.

plot_residuals	<i>Diagnostic residual plots for a BIC model</i>
----------------	--

Description

Diagnostic residual plots for a BIC model

Usage

```
plot_residuals(object, label_top = 3L)
```

Arguments

object	A fitted BEZI/BEOI gamlss model.
label_top	Integer; number of extreme points to label.

Value

Invisibly returns a list of residual vectors.

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